

Problem 1

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Problem 1 State the form of the limit. Determine whether the form is determinate or indeterminate. Evaluate each limit.

$$(a) \lim_{x \rightarrow 0} \frac{\sin(x) - \cos(x) + 1}{x^2 - x}$$

Explanation. The form is $\frac{0}{0}$; indeterminate. we can apply L.R.

$$\lim_{x \rightarrow 0} \frac{\sin(x) - \cos(x) + 1}{x^2 - x} = \lim_{x \rightarrow 0} \frac{\cos(x) + \sin(x)}{2x - 1} = \frac{\cos(0) + \sin(0)}{2(0) - 1} = \frac{1}{-1} = -1$$

$$(b) \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$$

Explanation. The form is $\frac{0}{0}$; indeterminate.

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x}, \text{ by L.R.}$$

The last limit also has the form $\frac{0}{0}$, so, we can apply L.R. again.

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

$$(c) \lim_{x \rightarrow \infty} \frac{e^x}{x^4}$$

Explanation. The form is $\frac{\infty}{\infty}$; indeterminate.

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^4} = \lim_{x \rightarrow \infty} \frac{e^x}{4x^3}, \text{ by L.R.}$$

The last limit also has the form $\frac{\infty}{\infty}$. It turns out that we can apply L.R. again, and again, and again:

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^4} = \lim_{x \rightarrow \infty} \frac{e^x}{4x^3} = \lim_{x \rightarrow \infty} \frac{e^x}{12x^2} = \lim_{x \rightarrow \infty} \frac{e^x}{24x} = \lim_{x \rightarrow \infty} \frac{e^x}{24} = \infty$$

This result shows that the function e^x grows much faster than x^4 , as x goes to ∞ .